



Division of Mathematical Statistics

KTH Matematik

MAKEUP EXAM IN SF1930 STATISTICAL LEARNING AND DATA ANALYSIS
FRIDAY DECEMBER 22 2021, 08:00–13:00

Examiner: Joakim Andén, 08-790 80 52.

Allowed aids: Pocket calculator, formula sheet (without notes), statistical tables.

The exam consists of two parts, denoted part I and part II. Part I consists of problems 1–12. In this part, you only need to provide an answer, either in the form of a numerical value with at least *three significant digits* ("värdesiffror") or in the form of a choice between different possible answers. Students who passed the midterm exam ("kontrollskrivning") do not have to answer problems 1–3 and will receive full credit for them (this will be possible for the regular exam and the makeup exam of the same year only). The preliminary limit is 9 points for a passing grade (E) and 8 points for the possibility to submit a supplementary assignment to upgrade to a passing grade (Fx).

Part II consists of problems 13–16 and each correct solution gives 10 points. This part will only be graded for students who have passed part I or have passed the supplementary assignment to upgrade to a passing grade. The points on part II are used to obtain a grade higher than E. In this part, arguments and computations should be detailed and well-motivated so that they are easy to follow. Any notation you introduce must be explained and defined. Numerical answers must be provided with at least *three significant digits*. Students who passed the homework assignment will receive up to 5 bonus points on part II (again, this applies for the regular exam and the makeup exam of the same year only).

The exam will be graded within three working weeks from the exam date and will be available at the student office at least seven weeks after the exam date.

Part I

Problem 1

Let $\mathbf{X} = [X_1, X_2, X_3]^T \sim N(\boldsymbol{\mu}, \Sigma)$ where

$$\boldsymbol{\mu} = [2, 3, -2]^T \quad \text{and} \quad \Sigma = \begin{bmatrix} 11 & -5 & 5 \\ -5 & 12 & 0 \\ 5 & 0 & 3 \end{bmatrix}.$$

Calculate $E[(2X_1 - 2X_2 - X_3)^2]$.

Problem 2

Suppose a random sample $\mathbf{X}_1, \dots, \mathbf{X}_n$, where $\mathbf{X}_i = (X_{1,i}, X_{2,i})^T$ is drawn from a bivariate normal distribution $N(\mu, \Sigma)$, with

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \text{and} \quad \Sigma^{-1} = \begin{pmatrix} k_{1,1} & k_{1,2} \\ k_{1,2} & k_{2,2} \end{pmatrix}.$$

Which of the following is a sufficient statistic for $(\mu_1, \mu_2, k_{1,1}, k_{1,2}, k_{2,2})$?

- (a) $(\prod_{i=1}^n X_{1,i}, \prod_{i=1}^n X_{2,i}, \prod_{i=1}^n X_{1,i}X_{2,i}),$
- (b) $(\sum_{i=1}^n X_{1,i}, \sum_{i=1}^n X_{2,i}),$
- (c) $(\sum_{i=1}^n X_{1,i}, \sum_{i=1}^n X_{2,i}, -\sum_{i=1}^n X_{1,i}^2/2, -\sum_{i=1}^n X_{2,i}^2/2, -\sum_{i=1}^n X_{1,i}X_{2,i}),$
- (d) $(\sum_{i=1}^n X_{1,i}, \sum_{i=1}^n X_{2,i}, \sum_{i=1}^n X_{1,i}^2, \sum_{i=1}^n X_{2,i}^2, (\sum_{i=1}^n X_{1,i}X_{2,i})^2 - (\sum_{i=1}^n X_{1,i})(\sum_{i=1}^n X_{2,i})).$

Problem 3

Let $X_1, X_2, \dots, X_n$ be an iid sample from a distribution with pdf

$$f(x | \lambda) = \begin{cases} \lambda x^{\lambda-1}, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate the method of moments estimator for λ given the data 0.61, 0.95, 0.57, 0.81.

Problem 4

Suppose we are given the data 2.55, 5.78, 4.60, 2.07 drawn from an $\text{exponential}(\lambda)$ distribution (with the rate parametrization) where $\lambda \sim \text{gamma}(3, 2)$ (also with the rate parametrization). Assuming the samples are conditionally independent given λ , what is the posterior expectation of λ given the data?

Problem 5

Suppose x is a realization of X is distributed as $\text{Poisson}(\lambda)$ for some parameter $\lambda > 0$. We want to test $H_0 : 0 < \lambda \leq \lambda_0$ against $H_1 : \lambda > \lambda_0$. Which of the following constitute an LRT for H_0 against H_1 , assuming c has been fixed to give the desired level α ?

- (a) Reject H_0 if $x > \lambda_0$ and $\left(\frac{\lambda_0}{x}\right)^x e^x \leq c.$
- (b) Reject H_0 if $x > \lambda_0$ and $\left(\frac{\lambda_0}{x}\right)^x e^{x-\lambda_0} \geq c.$
- (c) Reject H_0 if $x > \lambda_0$ and $\frac{\lambda_0^x}{x!} e^{-\lambda_0} \leq c.$
- (d) Reject H_0 if $x > \lambda_0$ and $\frac{x^x}{x!} e^{-x} \geq c.$

Problem 6

We model the monthly salary of Stockholm residents X as normally distributed with variance 20000 kr² and unknown mean μ (i.e., $X|\mu \sim N(\mu, 20000)$), and we assume $\mu \sim N(49200, 10000)$. A sample of individual monthly salaries (assumed to be conditionally independent given μ) is drawn:

47 600, 51 350, 50 500, 45 700, 51 400.

Let a_0 denote the action “decide $\mu \leq 49\,350$ ” and a_1 denote the action “decide $\mu > 49\,350$.” Conduct a Bayes test with a 0–1 loss function for the given data. Give the ratio of the posterior expected losses of a_0 and a_1 and state which action you choose based on the test.

Problem 7

Let X_1, X_2, X_3, X_4 be an iid sample from some distribution parameterized by $\theta > 0$. Suppose that for $H_0 : \theta \geq \theta_0$ versus $H_1 : 0 < \theta < \theta_0$, we have the level 10% test

Reject H_0 if $\max(x_1, x_2, x_3, x_4) < 0.56\theta_0$.

Compute a 90% confidence interval for θ given the data

23.6, 23.3, 27.5, 20.6.

What is the upper limit of this confidence interval?

Problem 8

Suppose we have the following samples, conditionally independent given λ , drawn from an exponential(λ) distribution, which we parametrize via rate:

2.28, 1.48, 2.93, 2.22, 0.09.

Assuming a gamma(2, 3) prior (with a rate parametrization), give the lower limit of the equal-tailed (two-sided) 95% credible interval for λ .

Problem 9

Let $X_1, X_2, \dots, X_n$ be iid samples from a Poisson(λ) distribution. Assuming the data is generated with $\lambda = 4$ and we sample obtain $n = 16$ data points, what is the smallest possible variance of an unbiased estimator of λ given this sample?

Problem 10

Let $(X_1, \dots, X_n)$ be jointly distributed Bernoulli random variables satisfying

$$P(X_1 = 1) = p_1,$$

$$P(X_i = 1 \mid X_1 = x_1, \dots, X_{i-1} = x_{i-1}) = \begin{cases} p_{11} & \text{if } x_{i-1} = 1, \\ p_{01} & \text{if } x_{i-1} = 0, \end{cases} \quad \text{for } i = 2, \dots, n,$$

where $p_1, p_{11}, p_{01} \in (0, 1)$. Compute the AIC of the model for $n = 3$ with the iid data

$$(1, 0, 1), \quad (1, 1, 1), \quad (0, 1, 1), \quad (0, 0, 1).$$

Problem 11

Suppose we were given the data $\mathbf{x} = (x_1, x_2, x_3) = (3.72, 1.52, 0.79)$ which we chose to model as three iid draws from a normal model $N(\mu, 1)$ with a normal prior $\mu \sim N(\theta, \sigma)$, where θ and σ are known. We then drew one sample $\tilde{\mu} = 1.95$ from our posterior $\mu \mid \mathbf{x}$, based on which we drew five simulated random samples of size three $(\tilde{x}_1, \tilde{x}_2, \tilde{x}_3)$ from the posterior predictive distribution $\tilde{X}_1, \tilde{X}_2, \tilde{X}_3 \mid X_1, X_2, X_3$. The samples are:

$$\begin{aligned} &(2.69, 1.68, 2.92) \\ &(1.31, 1.71, 2.70) \\ &(1.84, 4.84, 2.92) \\ &(1.14, 5.69, 1.78) \\ &(1.72, 3.11, 1.42) \end{aligned}$$

Hence, each pair $(\tilde{x}_1, \tilde{x}_2, \tilde{x}_3), \tilde{\mu}$ is a random sample from $(\tilde{X}_1, \tilde{X}_2, \tilde{X}_3, \mu) \mid \mathbf{x}$. Using these random samples, estimate the Bayesian p-value of the test quantity $T(X_1, X_2, X_3, \mu) = |\bar{X} - \mu|$ and state whether or not you reject the chosen model as a good fit for the data based on this posterior predictive check.

Problem 12

Suppose that X has pdf

$$f(x) = \begin{cases} xe^{-x^2/2}, & x \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

Provide a formula for generating a sample x of X using a sample u of $U \sim \text{uniform}(0, 1)$.

Part II

Problem 13

Independence of transformations of normal random variables. Let $X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$ be independent normally distributed random variables.

- (a) Derive the distribution of $X + Y$. (2 p)
- (b) Derive the distribution of $X - Y$. (2 p)
- (c) Give a necessary and sufficient condition on σ_X and σ_Y for $X + Y \perp X - Y$ to hold. (6 p)

Problem 14

The worried parent. The parent of a small child is concerned about traffic accidents at a nearby crossing. Before letting the child walk to school by themselves, she wants to make sure that the frequency of accidents is low. She supposes that the number of accidents (defined here as a car hitting a person crossing the street) during n hours is distributed as a Poisson random variable with parameter $n\mu$, where μ is the average number of accidents per hour. Being a slightly worried person, the parent assumes that accidents happen with relatively high frequency of about one every hundred hours. More specifically, she believes μ to be centered around 0.01 with a standard deviation of 0.005.

To allay her worries, she decides to observe the crossing in question for a number of hours and count the number of accidents. Assuming that she never does observe an accident, how many hours will she have to wait before she can conclude that the expected μ lies below 0.001? (10 p)

Problem 15

Effect evaluation via empirical Bayes. A laboratory conducted three trials on the effectiveness of a particular drug at treating an illness. Each trial had a control group, consisting of people that did not receive the drug, and a treatment group, consisting of people that received the drug. The number of participants in each group who recovered from the illness within two weeks of the administration of the drug to the treatment group was reported:

Trial	Control group size	Treatment group size	Control recoveries	Treatment recoveries
1	22	21	12	14
2	25	26	14	17
3	23	19	12	12

The recovery of individuals within a given group are considered to be independent and identically distributed. Let $\theta_{i,C}$ and $\theta_{i,T}$ denote the probability that a given participant in the i^{th} control group and i^{th} experimental group, respectively, recovers and assume $\theta_{1,C}, \theta_{2,C}, \theta_{3,C}$ and $\theta_{1,T}, \theta_{2,T}, \theta_{3,T}$ are random samples drawn, respectively, from a $\theta_C \sim \text{beta}(\alpha_C, \beta_C)$ and a $\theta_T \sim \text{beta}(\alpha_T, \beta_T)$ population.

- (a) Using maximum likelihood estimates to approximate the necessary θ 's, give a method of moments estimate for $\alpha_C, \beta_C, \alpha_T$, and β_T based on the available data. (4 p)
- (b) A large trial was then conducted with the following data recorded:

Trial	Control group size	Treatment group size	Control recoveries	Treatment recoveries
4	350	400	170	245

Using an empirical Bayes model, compute the posterior expectations of θ_C and θ_T given this new data $\mathbf{x}_4$. (4 p)

- (c) Give an approximate equal-tailed credible interval for $\theta_T - \theta_C$ using normal approximation to $\theta_T \mid \mathbf{x}_4$ and $\theta_C \mid \mathbf{x}_4$ and treating these two posteriors as independent. Can we conclude with 95% confidence that the drug is an effective treatment for the illness? Why or why not? (2 p)

Problem 16

Inference for the Weibull distribution. Consider the Weibull distribution with parameters scale parameter $\theta > 0$ and shape parameter $k > 0$. Its pdf is given by

$$f(x \mid \theta, k) = \frac{k}{\theta} \left(\frac{x}{\theta} \right)^{k-1} e^{-(x/\theta)^k}.$$

- (a) If X has the above distribution, show that X^k is exponentially distributed with rate parameter θ^{-k} . (2 p)
- (b) Suppose that we have a random sample $X_1, X_2, \dots, X_n$ from a Weibull distribution with known k . Derive a one-dimensional sufficient statistic for θ . (2 p)
- (c) Calculate the MLE $\hat{\theta}$ for θ for known k given a random sample $X_1, X_2, \dots, X_n$. (3 p)
- (d) For $k = 0.7$ and $n = 476$, you obtain the estimate $\hat{\theta} = 9.4$. Calculate an approximate 95% confidence interval for θ . (3 p)

Good luck!



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SOLUTIONS TO EXAM IN SF1930 STATISTICAL LEARNING AND DATA ANALYSIS

Problem 1

Let $Y = 2X_1 - 2X_2 + X_3$. Then $Y = A\mathbf{X}$, where $A = [2, -2, 1]$. Consequently $Y \sim N(A\boldsymbol{\mu}, A\Sigma A^T)$, where

$$A\boldsymbol{\mu} = 2 \cdot 2 - 2 \cdot 3 + (-1) \cdot (-2) = 0$$

and

$$A\Sigma A^T = 2^2 \cdot 11 + 2 \cdot 2 \cdot (-2) \cdot (-5) + 2 \cdot 2 \cdot (-1) \cdot 5 + (-2)^2 \cdot 12 + 2 \cdot (-2) \cdot 1 \cdot 0 + 1^2 \cdot 3 = 115.$$

Since $E[Y^2] = \text{Var}[Y] + E[Y]^2 = \text{Var}[Y] + 0^2$, we have $E[Y^2] = 115$.

Problem 2

The likelihood function of the data can be written as

$$\begin{aligned} f(\mathbf{x} \mid \mu_1, \mu_2, k_{1,1}, k_{1,2}, k_{2,2}) \\ &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi \det(\Sigma)}} \exp \left(-\frac{1}{2} (x_{1,i} - \mu_1 \quad x_{2,i} - \mu_2) \begin{pmatrix} k_{1,1} & k_{1,2} \\ k_{1,2} & k_{2,2} \end{pmatrix} \begin{pmatrix} x_{1,i} - \mu_1 \\ x_{2,i} - \mu_2 \end{pmatrix} \right), \\ &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi \det(\Sigma)}} \exp \left(-\frac{1}{2} \alpha_i \right), \end{aligned}$$

where, by doing the matrix multiplication, we see that

$$-\frac{1}{2} \alpha_i = k_{1,1} \left(\frac{-x_{1,i}^2}{2} \right) + k_{2,2} \left(\frac{-x_{2,i}^2}{2} \right) + k_{1,2} (-x_{1,i} x_{2,i}) + (k_{1,1} \mu_1 + k_{1,2} \mu_2) (x_{1,i}) + (k_{2,2} \mu_2 + k_{1,2} \mu_1) (x_{2,i}).$$

Hence, the answer is seen to be (c) by summing the values in the exponents in the product and applying the factorization theorem.

Problem 3

The first moment is given by

$$E[X \mid \lambda] = \int_0^1 x \cdot \lambda x^{\lambda-1} dx = \frac{\lambda}{\lambda + 1}.$$

(We can also get this by noting that $X \sim \text{beta}(\lambda, 1)$ and using the formula from the formula sheet.) We thus have the equation

$$\bar{x} = \frac{\lambda}{\lambda + 1},$$

which gives the solution

$$\lambda = \frac{\bar{x}}{1 - \bar{x}}.$$

Since the sample mean is

$$\bar{x} = \frac{1}{4}(0.61 + 0.95 + 0.57 + 0.81) = 0.735,$$

this gives the method of moments estimator $\lambda \approx 2.774$.

Problem 4

Since $X \sim \text{exponential}(\lambda)$ and $\lambda \sim \text{gamma}(3, 2)$, we know that $\lambda \mid \mathbf{x} \sim \text{gamma}(3 + n, 2 + \sum_i x_i)$. In our case, $n = 4$ and $\sum_i x_i = 2.55 + 5.78 + 4.60 + 2.07 = 15$. Hence, $\lambda \mid \mathbf{x} \sim \text{gamma}(7, 17)$. Since the gamma distribution is parametrized with the rate parameter, we know that

$$E[\lambda \mid \mathbf{x}] = 7/17 \approx 0.412.$$

Problem 5

Since $X \sim \text{Poisson}(\lambda)$, we have that

$$f(x) = \frac{\lambda^x}{x!} e^{-\lambda}.$$

We know that the likelihood $L(\lambda; x) = \lambda^x e^{-\lambda}/x!$ is maximized globally by $\lambda = x$. Consequently

$$\max_{\lambda} L(\lambda; x) = L(x; x) = \frac{x^x}{x!} e^{-x}.$$

Over the null hypothesis set $\Theta_0 = (0, \lambda_0]$, its maximum is $\lambda = x$ when $x \leq \lambda_0$, otherwise it is $\lambda = \lambda_0$, since the likelihood is strictly increasing for $\lambda < x$. We thus have

$$\max_{0 < \lambda \leq \lambda_0} L(\lambda; x) = \begin{cases} \frac{x^x}{x!} e^{-x}, & x \leq \lambda_0, \\ \frac{\lambda_0^x}{x!} e^{-\lambda_0}, & x > \lambda_0. \end{cases}$$

This gives the likelihood ratio

$$\ell(x) = \begin{cases} 1, & x \leq \lambda_0, \\ \frac{\lambda_0^x}{x^x} e^{-\lambda_0 + x}, & x > \lambda_0 \end{cases}$$

The LRT rejects H_0 if $\ell(x)$ is less than some constant in $(0, 1)$, which implies that that $x > \lambda_0$. Furthermore, when $x > \lambda_0$, we have that $\ell(x) = (\lambda_0/x)^x e^{-\lambda_0 + x}$. Requiring that this is smaller than some constant is equivalent to requiring that

$$\left(\frac{\lambda_0}{x}\right)^x e^x \leq c,$$

for some constant c . Consequently, the answer is (a).

Problem 6

Since $X \sim N(\mu, 20000)$ and $\mu \sim N(49200, 10000)$, we know from the formula sheet that $\mu \mid \mathbf{x} \sim N(\mu_n, \tau_n^2)$ where

$$\mu_n = \frac{\frac{49200}{10000} + \frac{5\bar{x}}{20000}}{\frac{1}{10000} + \frac{5}{20000}} \approx 49\,271.423$$

since $\bar{x} = 49\,300$, and

$$\tau_n^2 = \frac{(10000)(20000)}{20000 + (5)(10000)} \approx 2\,857.143.$$

Hence, $\tau_n \approx \sqrt{2\,857.143} \approx 53.452$. Since $\mu \mid \mathbf{x}$ is normally distributed, $\mu = (53.452)z + 49271.423$ for $Z \sim N(0, 1)$. Note that a_0 corresponds to $\mu \leq 49\,350$, or equivalently,

$$z \leq \frac{49350 - 49271.423}{53.452} \approx 1.47.$$

We want to pick the action a_0 or a_1 that minimizes the Bayes risk, or equivalently, that minimizes the posterior expected loss. For a_0 , the posterior expected loss is calculated as

$$\begin{aligned} \int_{-\infty}^{\infty} L(\mu, a_0) \pi(\mu \mid \mathbf{x}) d\mu &= \int_{49350}^{\infty} \pi(\mu \mid \mathbf{x}) d\mu, \\ &= \int_{1.47}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz, \\ &\approx 1 - 0.9292, \\ &\approx 0.071. \end{aligned}$$

It follows that the posterior expected loss of a_1 is approximately 0.929, and the ratio of the posterior expected losses is $0.071/0.929 \approx 0.076$. Hence, we decide on a_0 and accept the hypothesis that $\mu \leq 49\,350$.

Problem 7

The critical region of the test is

$$\{(x_1, x_2, x_3, x_4) : \max(x_1, x_2, x_3, x_4) < 0.56\theta_0\},$$

so the acceptance region is

$$A(\theta_0) = \{(x_1, x_2, x_3, x_4) : \max(x_1, x_2, x_3, x_4) \geq 0.56\theta_0\}.$$

Inverting the test, we obtain the confidence interval

$$C(x) = \{\theta : (x_1, x_2, x_3, x_4) \in A(\theta)\} = \{\theta : \max(x_1, x_2, x_3, x_4) \geq 0.56\theta\} = [0, \frac{1}{0.56} \max(x_1, x_2, x_3, x_4)].$$

In this case, we have $\max(x_1, x_2, x_3, x_4) = 27.5$, so the interval is $[0, 49.107]$. The answer is therefore 49.107.

Problem 8

Since $X \sim \text{exponential}(\lambda)$ with prior $\lambda \sim \text{gamma}(2, 3)$, we know that

$$\lambda \mid x_1, \dots, x_n \sim \text{gamma}(2 + n, 3 + \sum_i x_i).$$

In our case, $n = 5$ and

$$\sum_i x_i = 2.28 + 1.48 + 2.93 + 2.22 + 0.09 = 9.$$

So $\lambda \mid \mathbf{x} \sim \text{gamma}(7, 12)$. Recall that if $X \sim \text{gamma}(\alpha, \beta)$ then

$$2\beta X \sim \text{gamma}(\alpha, 1/2) = \chi^2(2\alpha).$$

Hence, $24\lambda \mid \mathbf{x} \sim \chi^2(14)$ in our case. Using the table for a χ^2 distribution, we calculate that an approximate 95% equal-tailed credible interval is

$$\left[\frac{1}{24} \chi^2_{1-0.025}(14), \frac{1}{24} \chi^2_{0.025}(14) \right].$$

The lower limit is given by

$$\frac{\chi^2_{0.975}(14)}{24} \approx \frac{5.63}{24} \approx 0.235.$$

Problem 9

We first compute the log-density

$$\log f(x; \lambda) = -\log x + \log \lambda x - \lambda,$$

then differentiate it twice with respect to λ to obtain

$$\frac{d^2}{d\lambda^2} \log f(x; \lambda) = -\frac{1}{\lambda^2} x.$$

Consequently, we have the information number

$$I(\lambda) = -n \mathbb{E} \left[\frac{d^2}{d\lambda^2} \log f(X; \lambda) \right] = -n \mathbb{E} \left[-\frac{1}{\lambda^2} X \right] = \frac{n}{\lambda}.$$

The Cramér–Rao lower bound thus gives us that, for any unbiased estimator $\hat{\lambda}$ of λ , we have

$$\text{Var}[\hat{\lambda}] \geq \frac{1}{I(\lambda)} = \frac{\lambda}{n},$$

which in our case is equal to $4/16 = 0.25$.

Problem 10

Since the variables are Bernoulli, the conditions defining the model also imply that

$$P(X_1 = 0) = 1 - p_1,$$

$$P(X_i = 0 \mid X_1, \dots, X_{i-1}) = \begin{cases} 1 - p_{11} & \text{if } X_{i-1} = 1, \\ 1 - p_{01} & \text{if } X_{i-1} = 0, \end{cases} \quad \text{for } i = 2, 3.$$

Combining this observation with the fact that

$$P(X_1, X_2, X_3) = P(X_1)P(X_2 \mid X_1)P(X_3 \mid X_1, X_2),$$

the likelihood of the iid data is seen to be

$$P(\mathbf{x} \mid p_1, p_{11}, p_{01}) = p_1^2(1 - p_1)^2 p_{11}^3(1 - p_{11})^1 p_{01}^3(1 - p_{01})^1.$$

It follows that the log-likelihood equations are

$$\begin{aligned}\frac{d}{dp_1} \log P(\mathbf{x} \mid p_1, p_{11}, p_{01}) &= \frac{2}{p_1} - \frac{2}{1-p_1}, \\ \frac{d}{dp_{11}} \log P(\mathbf{x} \mid p_1, p_{11}, p_{01}) &= \frac{3}{p_{11}} - \frac{1}{1-p_{11}}, \quad \text{and} \\ \frac{d}{dp_{01}} \log P(\mathbf{x} \mid p_1, p_{11}, p_{01}) &= \frac{3}{p_{01}} - \frac{1}{1-p_{01}}.\end{aligned}$$

Since each of the three log-likelihood equations is also the log-likelihood equation of a binomial(p_1), binomial(p_{11}) and binomial(p_{01}) random variable, respectively, we know that the MLE of the data is

$$(\hat{p}_1, \hat{p}_{11}, \hat{p}_{01}) = \left(\frac{2}{4}, \frac{3}{4}, \frac{3}{4} \right) = \left(\frac{1}{2}, \frac{3}{4}, \frac{3}{4} \right).$$

Since the model contains three parameters, p_1, p_{11} and p_{01} , it follows that its AIC score is

$$\begin{aligned}\text{AIC} &= -2 \log P(\mathbf{x} \mid \hat{p}_1, \hat{p}_{11}, \hat{p}_{01}) + 2(3), \\ &= -2 \log \left(\left(\frac{1}{2} \right)^4 \left(\frac{3}{4} \right)^6 \left(\frac{1}{4} \right)^2 \right) + 6, \\ &\approx 20.54.\end{aligned}$$

Problem 11

We estimate the Bayesian p -value

$$p_B(\mathbf{x}, \mu) = P(T(\tilde{\mathbf{x}}, \mu) \geq T(\mathbf{x}, \mu) \mid \mathbf{x})$$

as the proportion of simulated $\tilde{\mathbf{x}} \sim \tilde{\mu} \mid \mathbf{x}$ for which

$$T(\tilde{\mathbf{x}}, \tilde{\mu}) \geq T(\mathbf{x}, \tilde{\mu}).$$

For our data, we get

$$T(\mathbf{x}, \tilde{\mu}) = T(\mathbf{x}, 1.95) = |2.01 - 1.95| = 0.06.$$

For our simulated draws we get:

$$\begin{aligned}T((2.69, 1.68, 2.92), \tilde{\mu}) &= 0.48 \\ T((1.31, 1.71, 2.70), \tilde{\mu}) &= 0.04 \\ T((1.84, 4.84, 2.92), \tilde{\mu}) &= 1.25 \\ T((1.14, 5.69, 1.78), \tilde{\mu}) &= 0.92 \\ T((1.72, 3.11, 1.42), \tilde{\mu}) &= 0.13.\end{aligned}$$

Hence, $p_B \approx 4/5$. Based on this posterior predictive check, the data $\mathbf{x}$ is no more extreme than 4/5 of our simulated draws, so we do not reject the model.

Problem 12

First, we calculate the cumulative distribution function

$$F(x) = \int_0^x y e^{-y^2/2} dy.$$

With the substitution $t = y^2/2$, we have $dt = ydy$, so this gives

$$F(x) = \int_0^{x^2} e^{-t} dt = [-e^{-t}]_{t=0}^{x^2/2} = (-e^{-x^2/2} + 1).$$

Inverting $F(x) = u$, we obtain

$$x = \sqrt{-2 \log(1-u)},$$

which is the desired formula. (Another acceptable answer is $x = \sqrt{-2 \log(u)}$ since the uniform distribution is symmetric about $1/2$.)

Problem 13

a) We first note that since X and Y are independent, we have that their joint distribution is bivariate normal $(X, Y) \sim N(\mu, \Sigma)$, where

$$\mu = \begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \quad \text{and} \quad \Sigma = \begin{pmatrix} \sigma_X^2 & 0 \\ 0 & \sigma_Y^2 \end{pmatrix}.$$

If $U = X + Y$ and $V = X - Y$, we know that

$$\begin{pmatrix} U \\ V \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix}$$

It follows from a theorem in class that

$$\begin{pmatrix} \mu_{X+Y} \\ \mu_{X-Y} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix} = \begin{pmatrix} \mu_X + \mu_Y \\ \mu_X - \mu_Y \end{pmatrix}$$

and the covariance matrix of (U, V) is

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \Sigma \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} \sigma_X^2 + \sigma_Y^2 & \sigma_X^2 - \sigma_Y^2 \\ \sigma_X^2 - \sigma_Y^2 & \sigma_X^2 + \sigma_Y^2 \end{pmatrix}.$$

Since the distribution of (U, V) is bivariate normal with the above mean and covariance matrix, it follows that the marginal distributions of U and V are bivariate normal with $X + Y = U \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2)$ and $X - Y = V \sim N(\mu_X - \mu_Y, \sigma_X^2 + \sigma_Y^2)$. Hence the answer to (a) is $X + Y \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2)$.

b) The above analysis also showed that $X - Y \sim N(\mu_X - \mu_Y, \sigma_X^2 + \sigma_Y^2)$.

c) By a theorem in class, we know that for bivariate normal $(X, Y) \sim N(\mu, \Sigma)$, $\text{Cov}[X, Y] = 0$ if and only if $X \perp Y$. Hence, since the above shows that $(X + Y, X - Y) \sim N(\mu', \Sigma')$ where

$$\mu' = \begin{pmatrix} \mu_X + \mu_Y \\ \mu_X - \mu_Y \end{pmatrix} \quad \text{and} \quad \Sigma' = \begin{pmatrix} \sigma_X^2 + \sigma_Y^2 & \sigma_X^2 - \sigma_Y^2 \\ \sigma_X^2 - \sigma_Y^2 & \sigma_X^2 + \sigma_Y^2 \end{pmatrix},$$

we know that a necessary and sufficient condition for $X + Y \perp X - Y$ is $\sigma_X^2 = \sigma_Y^2$.

Problem 14

Since the observed variable is Poisson, a natural choice for a prior is the family of gamma distributions, since these form conjugate priors to the Poisson distribution. We thus see α_0, λ_0 such that

$$\frac{\alpha_0}{\lambda_0} = 0.01 \quad \text{and} \quad \frac{\alpha_0}{\lambda_0^2} = 0.05^2.$$

A solution is given by $\alpha_0 = 4$ and $\lambda_0 = 400$.

We thus have $\mu \sim \text{gamma}(\alpha_0, \lambda_0)$ with $X \mid \mu \sim \text{Poisson}(n\mu)$, which gives that $\mu \mid X \sim \text{gamma}(\alpha_0, \lambda_0 + n)$. To ensure that the posterior mean is less than 0.001, we require that

$$\frac{\alpha_0}{\lambda_0 + n} \leq 0.001,$$

which is equivalent to

$$n \geq \frac{\alpha_0}{0.001} - \lambda_0 = 4000 - 400 = 3600.$$

In other words, the parent would have to stand there for 3600 hours, or 150 days.

Problem 15

a) Using maximum likelihoods, we approximate the $\theta_{C,i}$'s as

$$\theta_{1,C} \approx \frac{12}{22}, \theta_{2,C} \approx \frac{14}{25}, \theta_{3,C} \approx \frac{12}{23},$$

from which it follows that

$$\begin{aligned} \frac{1}{3} \sum_{i=1}^2 X_{i,C} &= \frac{1}{3} \left(\frac{12}{22} + \frac{14}{25} + \frac{12}{23} \right) \approx 0.542, \quad \text{and} \\ \frac{1}{3} \sum_{i=1}^2 X_{i,C}^2 &= \frac{1}{3} \left(\left(\frac{12}{22} \right)^2 + \left(\frac{14}{25} \right)^2 + \left(\frac{12}{23} \right)^2 \right) \approx 0.294. \end{aligned}$$

Similarly, for the treatment groups, we get

$$\theta_{1,T} \approx \frac{14}{21}, \theta_{2,T} \approx \frac{17}{26}, \theta_{3,T} \approx \frac{12}{19},$$

and

$$\begin{aligned} \frac{1}{3} \sum_{i=1}^2 X_{i,T} &= \frac{1}{3} \left(\frac{14}{21} + \frac{17}{26} + \frac{12}{19} \right) \approx 0.651, \quad \text{and} \\ \frac{1}{3} \sum_{i=1}^2 X_{i,T}^2 &= \frac{1}{3} \left(\left(\frac{14}{21} \right)^2 + \left(\frac{17}{26} \right)^2 + \left(\frac{12}{19} \right)^2 \right) \approx 0.424. \end{aligned}$$

For the control group, we then find a method of moments estimator by solving the system of equations

$$0.542 = \frac{\alpha_C}{\alpha_C + \beta_C}, \quad \text{and} \quad 0.294 = \frac{\alpha_C(\alpha_C + 1)}{(\alpha_C + \beta_C)(\alpha_C + \beta_C + 1)},$$

which leads to

$$\alpha_C = \frac{1 - \frac{0.294}{0.542}}{\frac{0.294}{0.542^2} - 1} \approx 569.559, \quad \text{and}$$

$$\beta_C = \alpha_C \left(\frac{1}{0.542} - 1 \right) = 569.559 \left(\frac{1}{0.542} - 1 \right) \approx 481.279.$$

Similarly, for the treatment group, we get

$$\alpha_T = \frac{1 - \frac{0.424}{0.651}}{\frac{0.424}{0.651^2} - 1} \approx 742.598, \quad \text{and}$$

$$\beta_T = \alpha_T \left(\frac{1}{0.651} - 1 \right) = 742.598 \left(\frac{1}{0.651} - 1 \right) \approx 398.106.$$

b) Since we are using an empirical Bayes model, we can take our point estimates for $\alpha_C, \beta_C, \alpha_T$, and β_T and assume that our prior distributions are $\theta_C \sim \text{beta}(569.559, 481.279)$ and $\theta_T \sim \text{beta}(742.598, 398.106)$. Since the data is binomial and the beta family is conjugate to the binomial family, we know that

$$\theta_C \mid \mathbf{x}_4 \sim \text{beta}(569.559 + 170, 481.279 + 180), \quad \text{and}$$

$$\theta_T \mid \mathbf{x}_4 \sim \text{beta}(742.598 + 245, 398.106 + 155).$$

Hence,

$$E[\theta_C \mid \mathbf{x}_4] \approx \frac{569.559 + 170}{569.559 + 170 + 481.279 + 180} \approx 0.528, \quad \text{and}$$

$$E[\theta_T \mid \mathbf{x}_4] \approx \frac{742.598 + 245}{742.598 + 245 + 398.106 + 155} \approx 0.641.$$

c) Since our posterior distribution for θ_T is beta, we know that the posterior variance is

$$\frac{(742.598 + 245)(398.106 + 155)}{(742.598 + 398.106 + 400)^2(742.598 + 398.106 + 400 + 1)} \approx 0.0001.$$

Similarly, the posterior variance of θ_C is approximately 0.0002. Approximating $\theta_C \mid \mathbf{x}_4$ and $\theta_T \mid \mathbf{x}_4$ with independent normals $N(0.641, 0.0001)$ and $N(0.528, 0.0002)$, respectively, we have that $\theta_T - \theta_C \mid \mathbf{x}_4$ is approximately $N(0.113, 0.0003)$. So we estimate a 95% equal-tailed credible interval to be

$$[(0.0173)(-1.96) + 0.113, (0.0173)(1.96) + 0.113] \approx [0.079, 0.147].$$

Since 0 lies to the left of this confidence interval, we can say with 95% confidence that the drug is effective.

Problem 16

- (a) For $g(x) = x^k$, we have the inverse $g^{-1}(y) = y^{1/k}$ with derivative $(dg^{-1}/dy)(y) = y^{1/k-1}/k$. Using the change-of-variables formula, we have the pdf for $Y = X^k$ given by

$$\begin{aligned} f_Y(y) &= f_X(g^{-1}(y)) \left| \frac{dg^{-1}}{dy}(y) \right| \\ &= \frac{k}{\theta} \left(\frac{y^{1/k}}{\theta} \right)^{k-1} e^{-(y^{1/k}/\theta)^k} \cdot \frac{1}{k} y^{1/k-1} \\ &= \theta^{-k} y^{1-1/k} e^{-\theta^{-k} y} \cdot y^{1/k-1} = \theta^{-k} e^{-\theta^{-k} y}. \end{aligned}$$

This is the pdf of an exponentially distributed random variable with rate parameter θ^{-k} .

- (b) First, we form the simultaneous pdf for the random sample $\mathbf{X} = [X_1, X_2, \dots, X_n]$, which gives

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{k^n}{\theta^n} \left(\frac{\prod_{i=1}^n x_i}{\theta^n} \right)^{k-1} e^{-\theta^{-k} \sum_{i=1}^n x_i^k}.$$

We can rewrite this as $f_{\mathbf{X}}(\mathbf{x}) = h(\mathbf{x})g(t(\mathbf{x})|\theta)$, where

$$h(\mathbf{x}) = k^n \prod_{i=1}^n x_i^{k-1},$$

$$g(t|\theta) = \theta^{-nk} e^{-\theta^{-k} t},$$

and finally

$$t(\mathbf{x}) = \sum_{i=1}^n x_i^k.$$

By the factorization theorem, $t(\mathbf{x})$ is a sufficient statistic.

- (c) Taking the log-likelihood, we obtain

$$\log L(\theta; t(\mathbf{x})) = -nk \log \theta - \theta^{-k} t(\mathbf{x}) + C,$$

where C is a constant that does not depend on θ . Differentiating with respect to θ now gives

$$\frac{d \log L(\theta; t(\mathbf{x}))}{d\theta} = -\frac{nk}{\theta} + k\theta^{-k-1} t(\mathbf{x}).$$

If we set this to zero and solve for θ , we obtain

$$\theta = \left(\frac{1}{n} t(\mathbf{x}) \right)^{1/k} = \left(\frac{1}{n} \sum_{i=1}^n x_i^k \right)^{1/k}.$$

To show that this is a global maximum, we compute the second derivative

$$\frac{d^2 \log L(\theta; t(\mathbf{x}))}{d\theta^2} = \frac{nk}{\theta^2} - k(k+1)\theta^{-k-2} t(\mathbf{x}) = \frac{k}{\theta^2} (n - (k+1)\theta^{-k} t(\mathbf{x}))$$

Plugging in $\theta^k = \frac{1}{n} t(\mathbf{x})$, we thus have

$$\frac{d^2 \log L(\theta; t(\mathbf{x}))}{d\theta^2} = \frac{k}{\theta^2} (n - (k+1)n) = -\frac{nk^2}{\theta^2} < 0,$$

so the log-likelihood is concave. Its sole critical point is therefore a global maximum.

- (d) At large n , the MLE approximately normally distributed. We have the (normalized) information number

$$\begin{aligned} J(\theta) &= -\mathbb{E} \left[-\frac{d^2}{d\theta^2} \log f(X|\lambda) \right] \\ &= -\mathbb{E} \left[\frac{k}{\theta^2} - \frac{k(k+1)}{\theta^{k+2}} X^k \right] \\ &= -\frac{k}{\theta^2} + \frac{k(k+1)}{\theta^{k+2}} \mathbb{E}[X^k] \\ &= -\frac{k}{\theta^2} + \frac{k(k+1)}{\theta^{k+2}} \theta^k = \frac{k^2}{\theta^2}, \end{aligned}$$

where in the last line we have used the fact that $X^k \sim \text{exponential}(\theta^{-k})$.

We thus have that the MLE $\hat{\theta}$ satisfies

$$\sqrt{n}(\hat{\theta} - \theta) \longrightarrow N(0, 1/J(\theta)) = N(0, \theta^2/k^2).$$

At large n , we then have approximately that $\hat{\theta}$ follows a $N(\theta, \theta^2/nk^2)$ distribution. Since θ is not known, we approximate the variance with $\hat{\theta}^2/nk^2$.

For the given parameters and observations, this gives the confidence interval

$$\hat{\theta} \pm \lambda_{0.025} \sqrt{\frac{\hat{\theta}^2}{nk^2}} \approx 9.2 \pm 1.96 \cdot 0.6155 \approx 9.2 \pm 1.2064 \approx [7.994, 10.406].$$